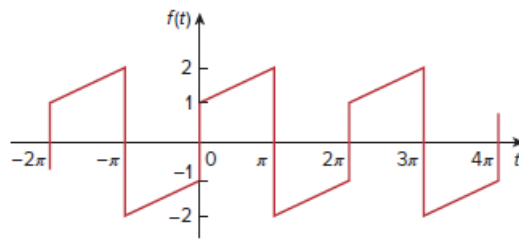


## Problem

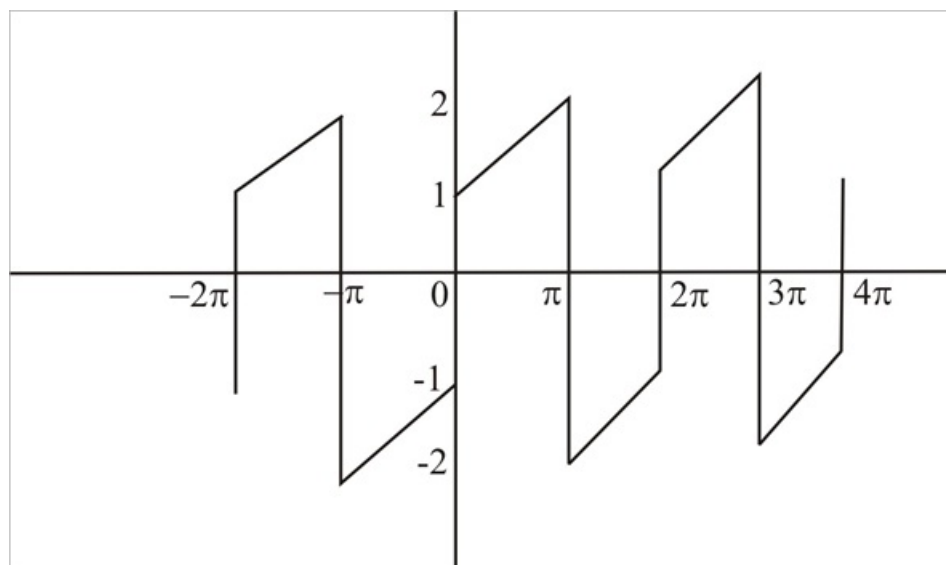
Obtain the complex Fourier coefficients of the signal in Fig. 17.62.

**Figure 17.62:**



## Step-by-step solution

## Step 1 of 2



## Step 2 of 2

$$f(t) = \begin{cases} \frac{t}{\pi} - 1 & -\pi < t < 0 \\ \frac{t}{\pi} + 1 & 0 < t < \pi \end{cases}$$

$$C_n = \frac{1}{T} \int_0^T f(t) e^{-jn\omega_0 t} dt$$

$$= \frac{1}{2\pi} \int_0^{2\pi} f(t) e^{-jnt} dt$$

$$= \frac{1}{2\pi} \left[ \int_{-\pi}^0 \left( \frac{t}{\pi} - 1 \right) e^{-jnt} dt + \int_0^{\pi} \left( \frac{t}{\pi} + 1 \right) e^{-jnt} dt \right]$$

$$C_n = \frac{1}{2\pi} \left[ \left( \frac{t}{\pi} - 1 \right) \frac{e^{-jnt}}{-jnt} - \int \frac{e^{-jnt}}{-\pi jnt} dt + \left( \frac{t}{\pi} + 1 \right) \frac{e^{-jnt}}{-jnt} - \int \frac{e^{-jnt}}{-jnt} dt \right]$$

$$= \frac{1}{2\pi} \left[ \left( \frac{t}{\pi} - 1 \right) \frac{e^{-jnt}}{-jnt} - \frac{e^{-jnt}}{-\pi jnt} \right]_{-\pi}^0 + \left( \frac{t}{\pi} + 1 \right) \frac{e^{-jnt}}{-jnt} - \frac{e^{-jnt}}{\pi(-jnt)^2} \right]_0^{\pi}$$

$$= \frac{1}{2\pi} \left[ \frac{1}{jn} + \frac{1}{n^2\pi} + \frac{2e^{-jn\pi}}{-jn} + \frac{e^{-jn\pi}}{n^2\pi} + \frac{2e^{-jn\pi}}{-jn} + \frac{e^{-jn\pi}}{n^2\pi} + \frac{1}{jn} + \frac{1}{n^2\pi} \right]$$

$$C_n = \frac{1}{2\pi} \left[ \frac{2}{jn} + \frac{4e^{-jn\pi}}{-jn} + \frac{e^{-jn\pi}}{-n^2\pi} + \frac{e^{-jn\pi}}{-n^2\pi} \right]$$

$$e^{-jn\pi} = \cos n\pi - j \sin n\pi \Rightarrow \cos n\pi (-1)^n$$

$$e^{jn\pi} = (-1)^n$$

$$C_n = \frac{1}{2\pi} \left[ \frac{2}{jn} + 4e^{-jn\pi} - \frac{4(-1)^n}{-jn} + \frac{(-1)^n}{jn} + \frac{(-1)^n}{n^2\pi} \right]$$

$$= \frac{1 - 2(-1)^n}{jn\pi}$$

$$f(t) = \sum_{n=-\infty}^{\infty} \frac{1 - 2(-1)^n}{jn\pi} e^{jn\pi t}$$